CHG 569
Environmental Pollution Engineering

Module V-3

Air Pollution Control DEVICES for PARTICULATES

The conventional devices used for controlling particulate emissions are:

- Wet collectors - Venturi scrubbers
- Centrifugal Separators -Cyclones
- Gravity settling chambers
- Electrostatic precipitators, and
- Fabric filters.

Wet collectors

These control devices remove particulate matter from gas streams by incorporating the particles into liquid droplets directly on contact

Mechanism: Inertial impaction / gravitational settling

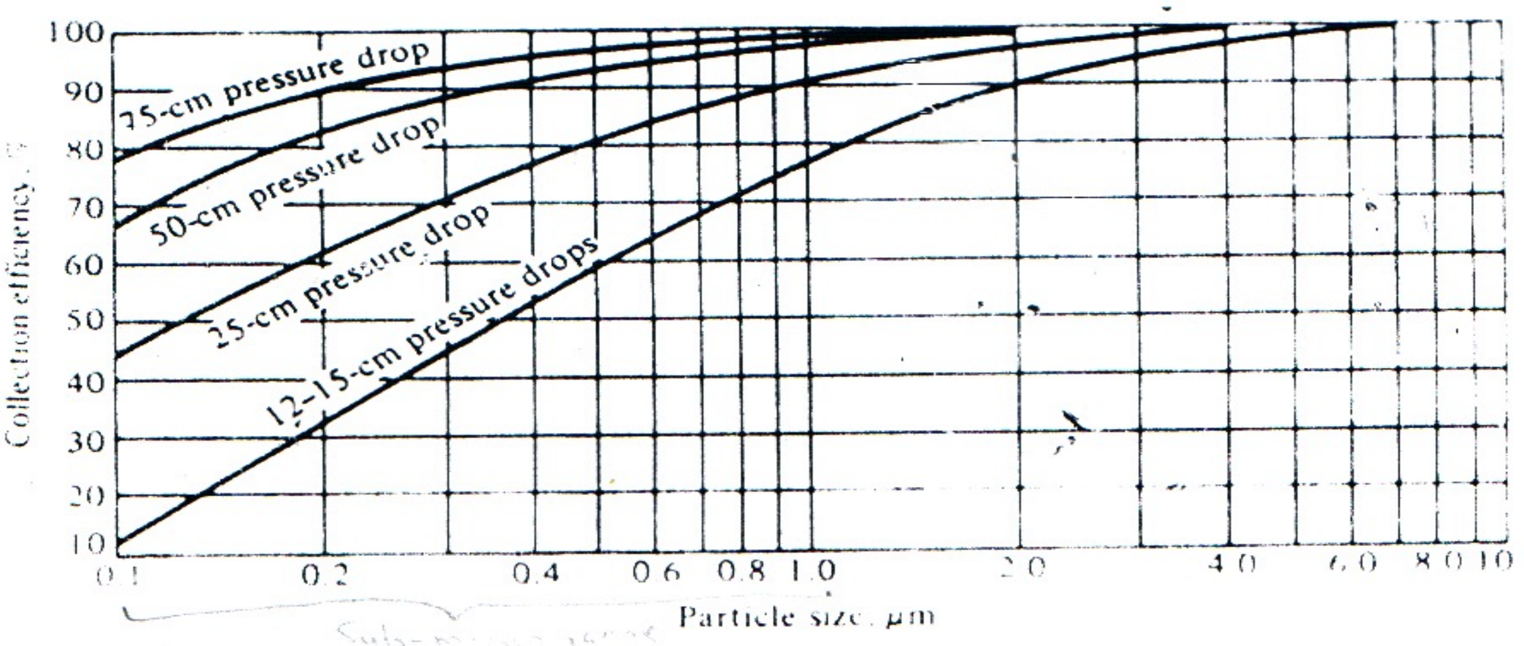
Examples of a wet scrubbers are:

- **Venturi scrubbers –** most popular 12 – 250 cm w. g
- Spray towers (with and without a bed) 1 – 4 cm w. g.
- Cyclone scrubbers 5 – 15 w.g.

Collection efficiency for well designed scrubbers is a function of the **energy consumed** in the air-to-water contact process, which is directly proportional to the **pressure drop - the contact-power theory**

Wet collectors - contd.

Contact-Power theory - "When compared at the same power consumption, all scrubbers give substantially the same degree of collection of a given dispersed dust, regardless of the mechanism involved and regardless of whether the pressure drop is obtained by high gas flow rates or high water flow rates." (Lapple and Kamack 1955)

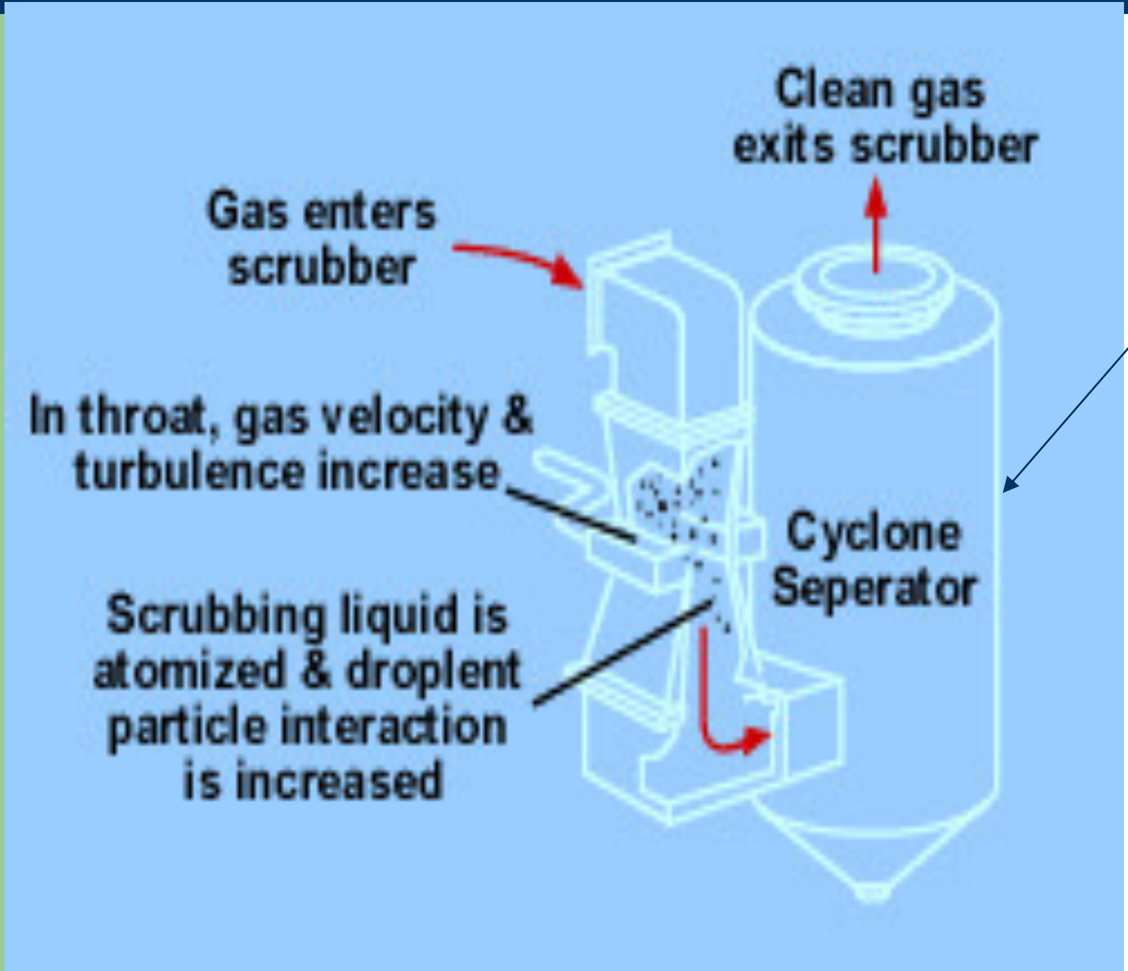


Graph of efficiency versus particle size for a wet scrubber (*From Shaheen [9-42]*) operating at various pressure drops.

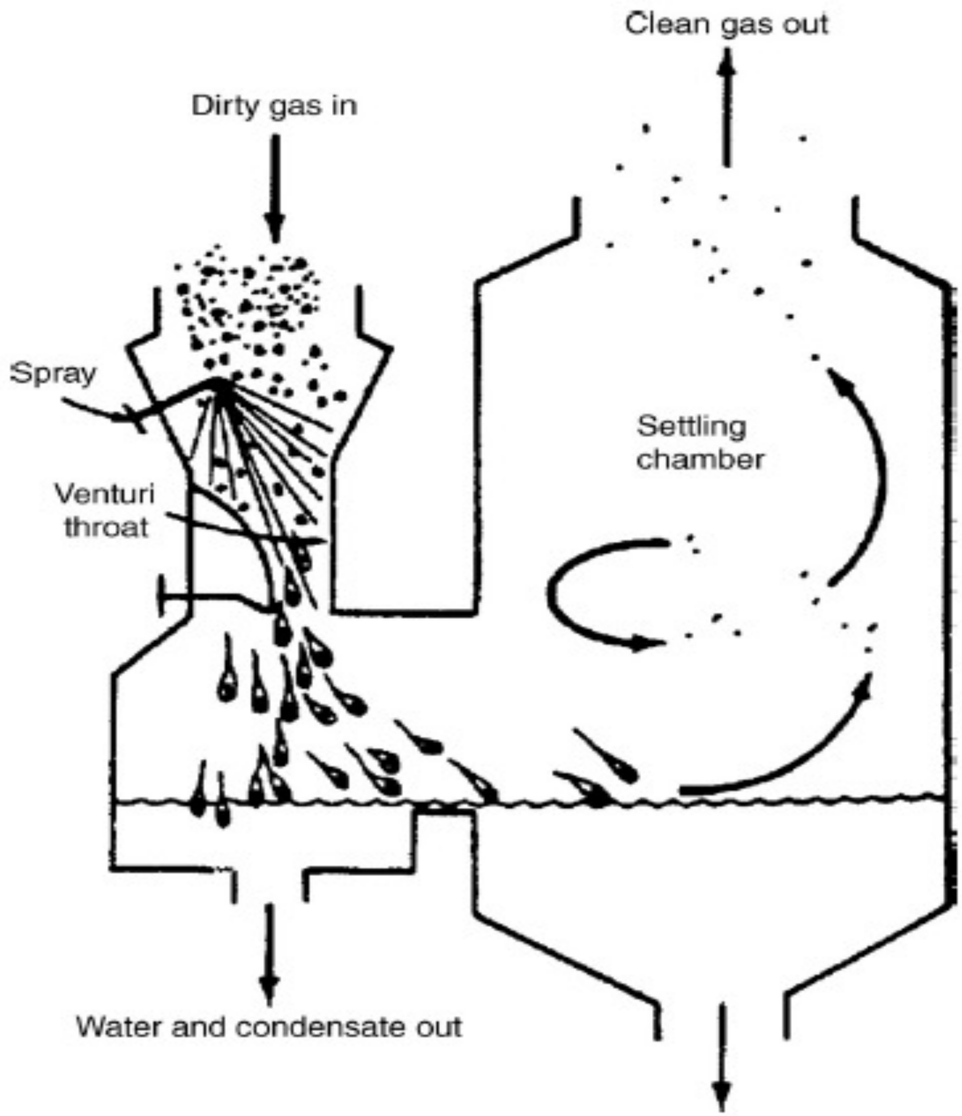
Efficiency not only increases with pressure drop but also with particle size

Wet collectors - contd.

Venturi Scrubber → most efficient for particle size 0.5 - 5μm
→ power cost relatively high bcos of the high ΔP caused by the high gas velocity



mist eliminator is important to achieve high removal efficiencies

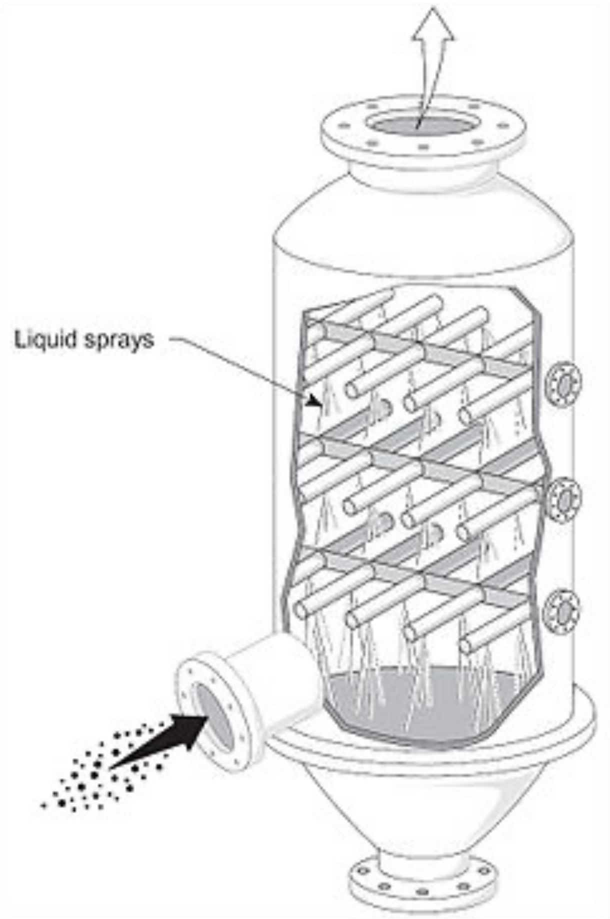

Clean gas out
Dirty gas in
Spray
Venturi throat
Settling chamber
Water and condensate out

Wet collectors - Contd

Spray tower scrubbers

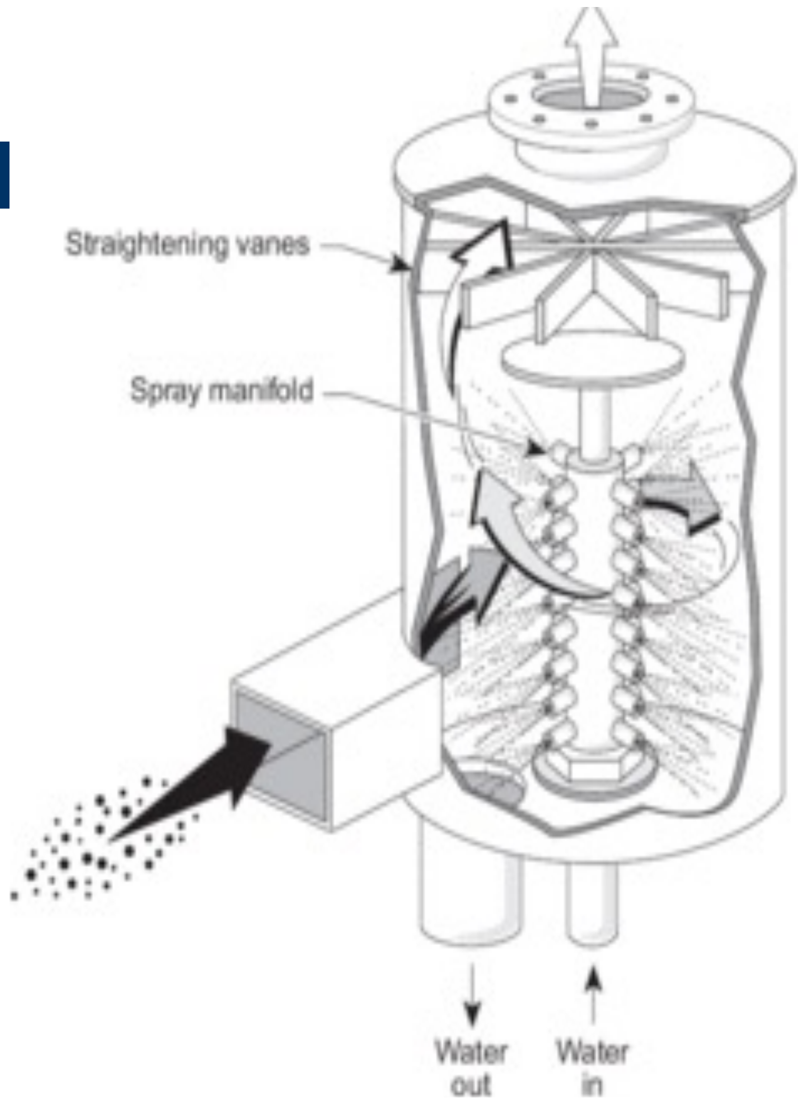
- low cost scrubbers, very little pressure drop
- Can handle both gaseous and particulate pollutants
- can handle large volume of gases

Effective for D_p > 10μm



Cyclone scrubbers

This basically is a cyclone or centrifugal separator with high pressure spray nozzle located within the cyclone chamber to generate fine spray (obstacles or targets for impaction)



Typical ranges for various wet scrubbers

Ranges of pressure drops and liquid-to-gas (L/G) ratios for various wet scrubbers				
Scrubber	Pressure drop, Δp		Liquid-to-gas ratio[1]	
	kPa	in. H_2O	L/m^3	gal/1000 ft^3
Venturi	1.5-25.0	5.0-100.0	0.4-5.0	3.0-40.0
Spray tower	0.12-0.75	0.5-3.0	0.7-2.7	5.0-20.0
Cyclonic spray	0.4-4.0	1.5-10.0	0.3-1.3	2.0-10.0

Predicting Scrubber (Venturi) efficiency

Several models are available for the calculation of Venturi particle collection efficiency, but this lecture is considering the **Johnstone equation** - assumes predominant mechanism of inertial impaction

Using the Johnstone equation: $\eta = 1 - \exp\left(-k\, {}^{L}\!/\!_{G}\right)\sqrt{k_p}$

where η = efficiency

k = correlation coefficient typically 0.1 to 0.2, 1000 acf/gal

${}^{L}\!/\!_{G}$ = liquid-to-gas ratio, gal/1000 acf

k_p = inertial impaction parameter, $= \left[\dfrac{D_p^2 \rho_p v_g C_s}{9 \mu_g D_c}\right]$

v_g = gas velocity at the throat, ft/sec

The collector (droplet, in this case) diameter (D_c) to be used in the impaction parameter is calculated in the following equation known as the Nukiyama -Tanasawa equation, given by:

$$D_c = \frac{50}{v_g} + 91.8\left({}^{L}\!/\!_{G}\right)^{1.5}$$

where

D_c = droplet diameter, cm

v_g = gas velocity in the throat, cm/s

${}^{L}\!/\!_{G}$ = liquid-to-gas ratio, dimensionless

Pressure drop can be derived from equations or performance curves. A simple approach for calculating pressure is

$$\Delta P = v_g^2 \,{}^{L}\!/_{G} \times 10^{-6}$$

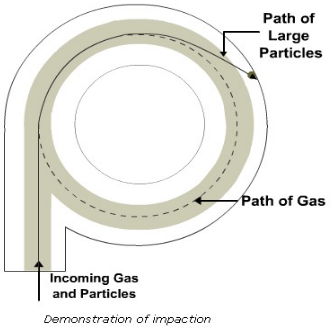
Where ΔP = pressure drop across venturi, cm H_2O

v_g = throat gas velocity, cm/sec

L/G = water to air volume ratio, L/m^3

Centrifugal Separators

Cyclones – the most common

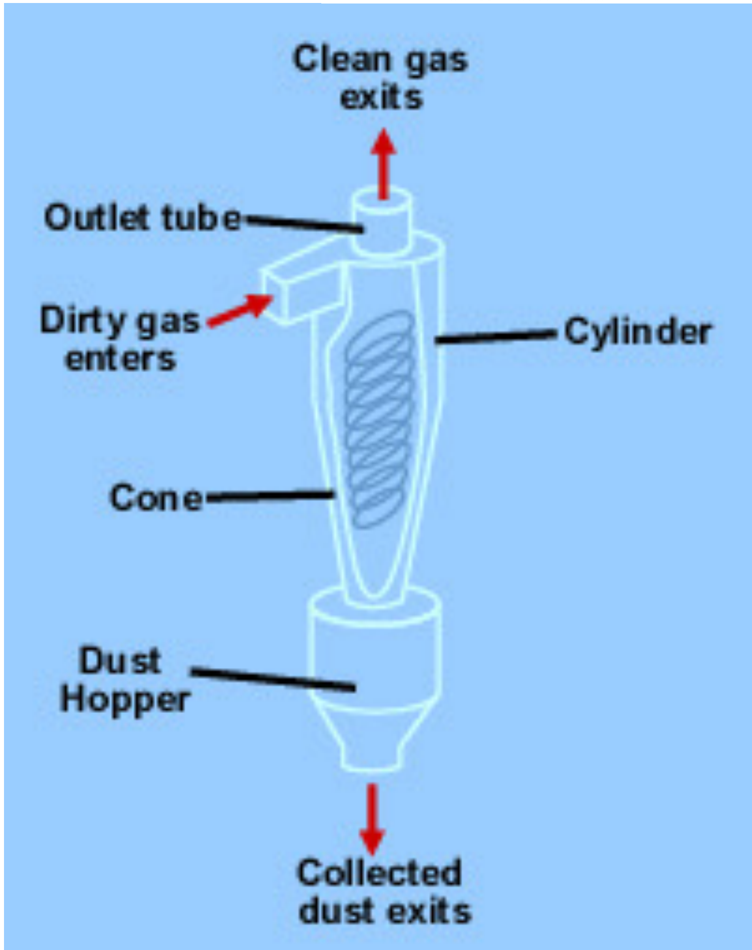


- Cyclones provide a low-cost, low-maintenance method of removing larger particulates from a gas stream

- The general principle of inertia separation is that the particulate-laden gas is forced to change direction. As gas changes direction, the inertia of the particles causes them to continue in the original direction and be separated from the gas stream.

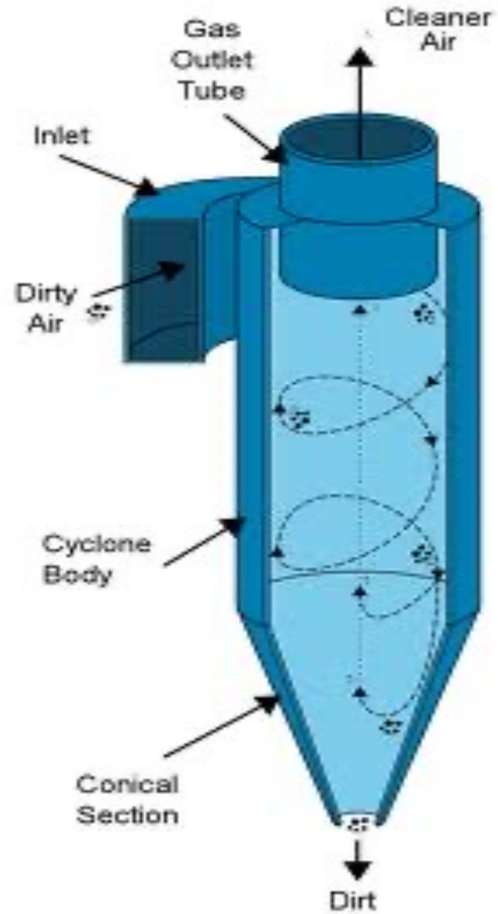
Effective for Dp > 10μm

For smaller particles use the high efficiency cyclone



Predicting cyclone efficiency

Several methods exist for evaluating the efficiency of a cyclone collector, but this lecture is considering the **Method of Lapple** – a relatively simple technique, one drawback is that it does not take the cyclone dimensions into consideration



Lapple Technique

Step 1: Calculate the cut diameter - particle size collected with 50% efficiency- from:

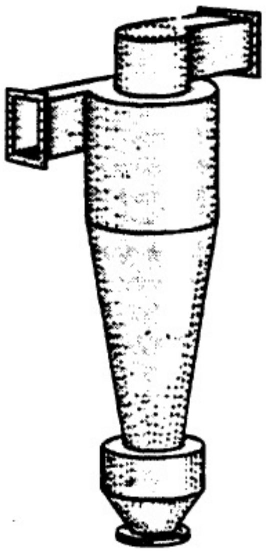
$$D_{p,\mathrm{cut}} = \left[\frac{9\mu_g W_i}{2\pi N v_i(\rho_p - \rho_g)}\right]^{0.5}$$

Where

- $D_{p,\,cut}$ = Particle cut size
- μ_g = gas viscosity
- W_i = width of rectangular cyclone inlet duct
- v_i = gas inlet velocity
- N = Number of turns = $\dfrac{v_i t}{\pi D}$

where D is the cyclone diameter, t the residence time, =

$$\frac{V_{\mathrm{cyclone}} - V_{\mathrm{outlet\ core}}}{Q}$$



Predicting cyclone efficiency - contd

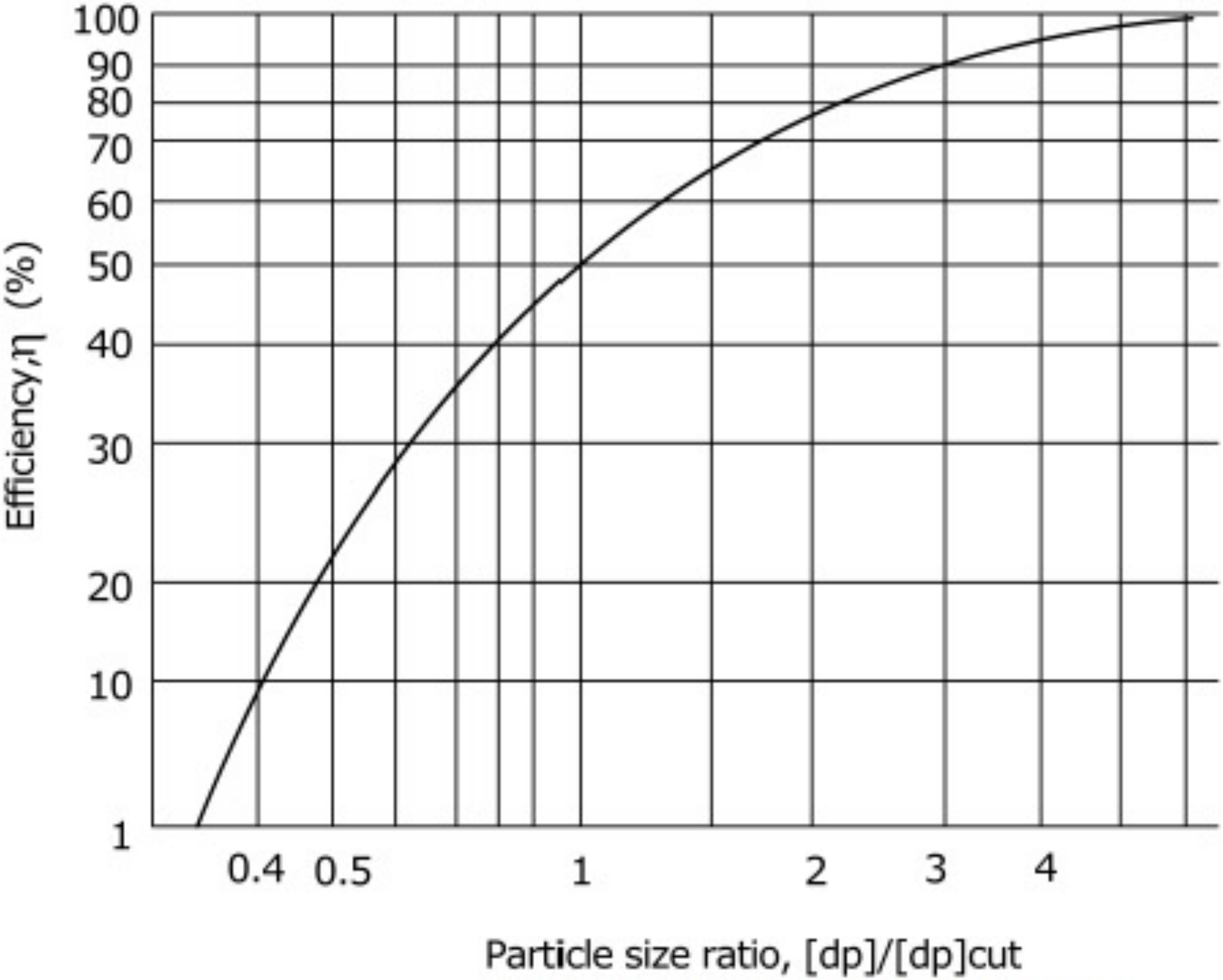
Step 2: Calculate the $D_p / D_{p,cut}$ ratio for each particle size of interest

Step 3: The fractional efficiencies, η_i, for each $D_p / D_{p,cut}$ ratio are read from the Figure in slide 15, or determined from:

$$\eta_i = \frac{1}{1 + \left(D_{p,cut} / D_{p_i} \right)^2}$$

Step 4: Determine overall efficiency from

$$\eta = \frac{\sum \eta_i m_i}{M}$$



Lapple cyclone efficiency curve

Predicting cyclone efficiency - contd

Since impaction relies on the flow velocity, a higher efficiency is the result of a higher flow velocity. However, a higher flow velocity is accompanied by a higher pressure drop across the system that can be translated into **cost** as a more powerful fan is required. The pressure drop can be determined by:

$$\Delta P = K_p \rho_g v_g^2$$

Where

ΔP = static pressure drop (in wc)

K_p = 0.013 to 0.024 (dimensionless)

ρ_g = gas density ($1b_m/ft^3$)

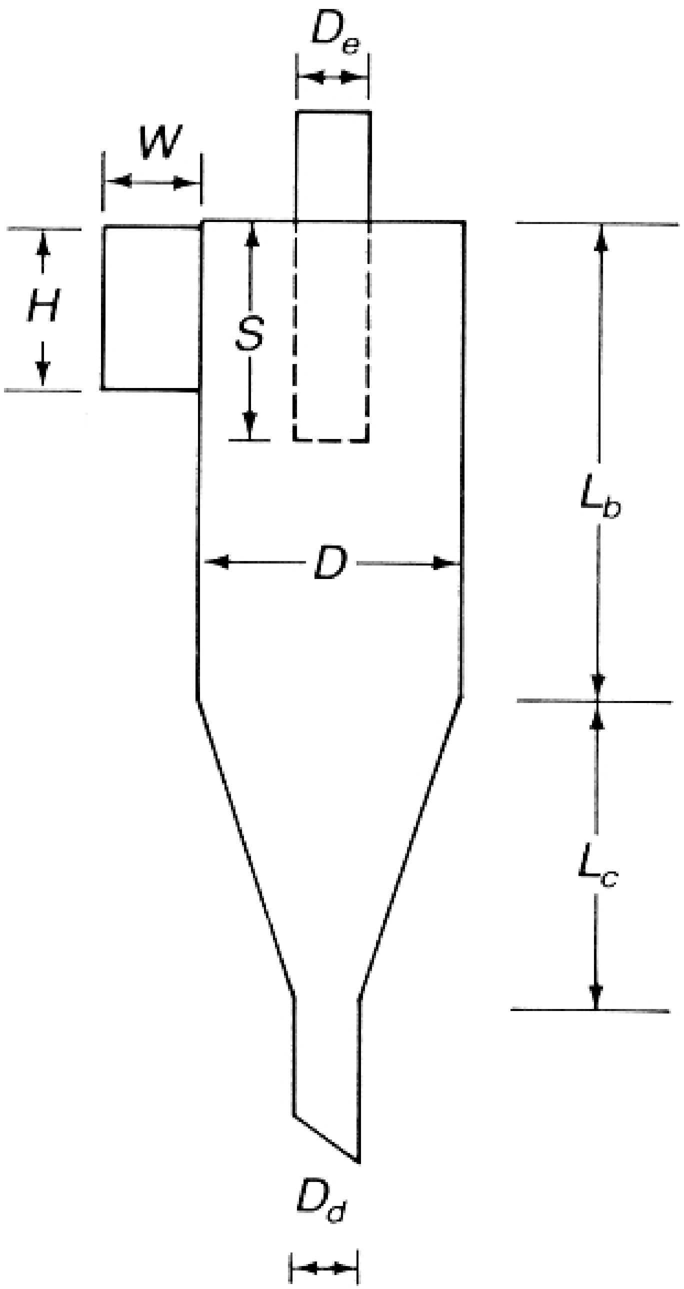
v_g = inlet velocity (ft/sec)

There are so many other correlations that can be used to determine pressure drop

Predicting cyclone efficiency - contd

Diameter of the smallest particle that can be completely removed, that is efficiency = 100% is estimated from:

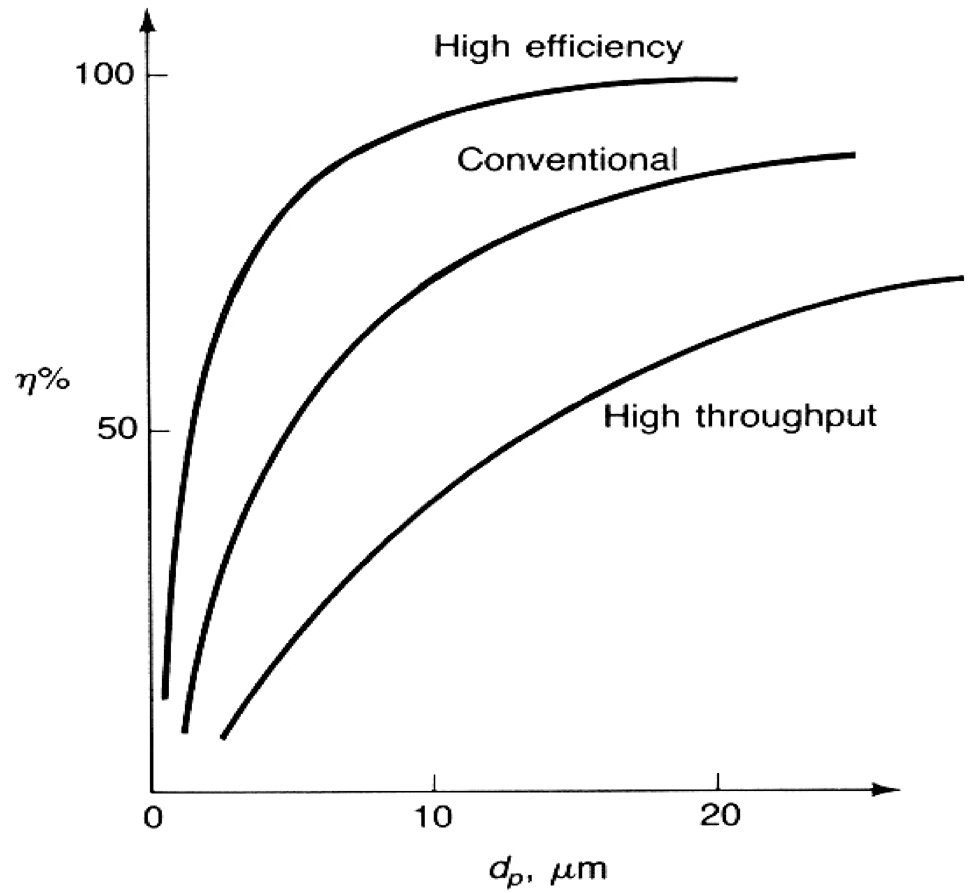
$$D_{p,\min} = \sqrt{\left[\frac{9\mu_g W_i}{\pi N v_i(\rho_p - \rho_g)}\right]}$$



Standard cyclone design dimensions

Cyclones are generally grouped into 3 classes:

- Conventional or general purpose.
- medium efficiency or high thruput
- high efficiency or medium thruput.



Standard cyclone design dimensions

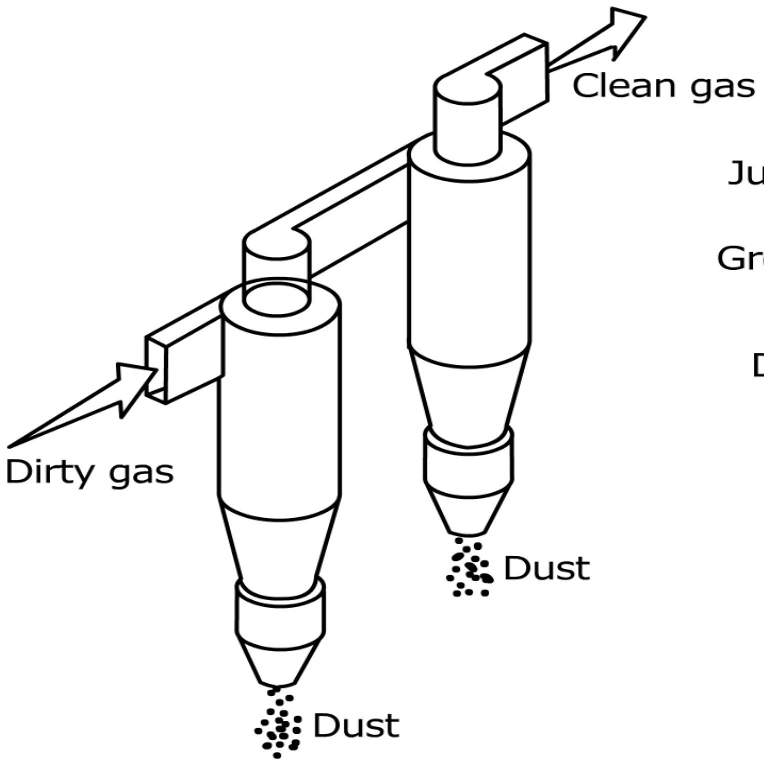
All dimensions listed are **normalized** to the cyclone's body diameter.

	Cyclone Type					
	High Efficiency		Conventional		High Throughput	
	(1)	(2)	(3)	(4)	(5)	(6)
Body Diameter, D/D	1.0	1.0	1.0	1.0	1.0	1.0
Height of Inlet, H/D	0.5	0.44	0.5	0.5	0.75	0.8
Width of Inlet, W/D	0.2	0.21	0.25	0.25	0.375	0.35
Diameter of Gas Exit, D_e/D	0.5	0.4	0.5	0.5	0.75	0.75
Length of Vortex Finder, S/D	0.5	0.5	0.625	0.6	0.875	0.85
Length of Body, L_b/D	1.5	1.4	2.0	1.75	1.5	1.7
Length of Cone, L_c/D	2.5	2.5	2.0	2.0	2.5	2.0
Diameter of Dust Outlet, D_d/D	0.375	0.4	0.25	0.4	0.375	0.4

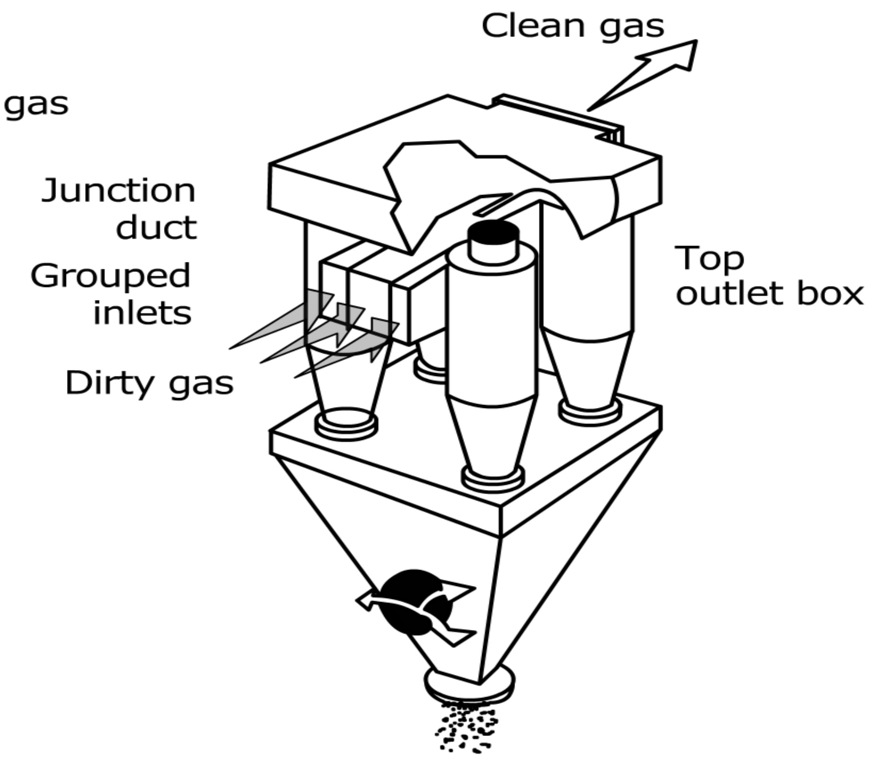
SOURCES: Columns (1) and (5) = Stairmand, 1951; columns (2), (4) and (6) = Swift, 1969; columns (3) = Lapple, 1951.

Arrangements of cyclone

Large diameter cyclones can be used in series or parallel arrangements in order to increase particulate matter removal efficiency or to increase gas flow capability.



A. two cyclones in series

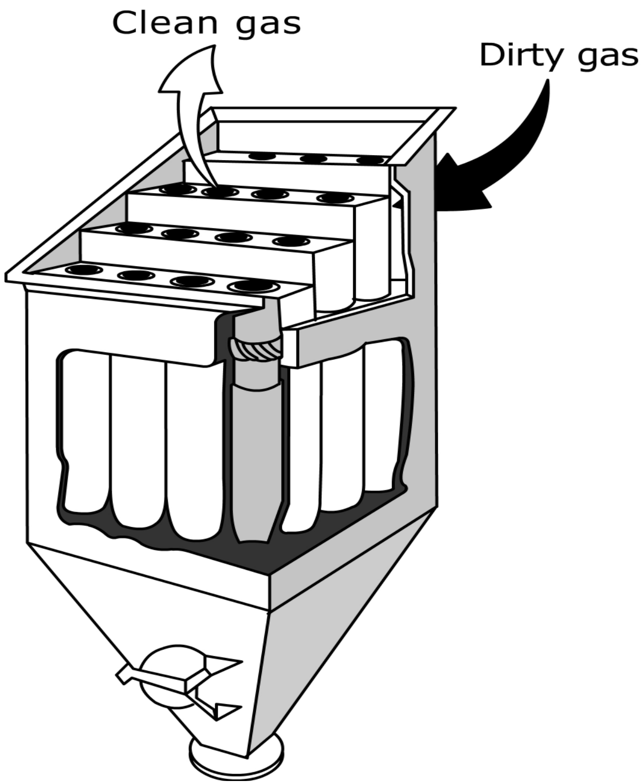


B. Four cyclones in parallel
(See next slide)

Multi- cyclone (Multiclones)

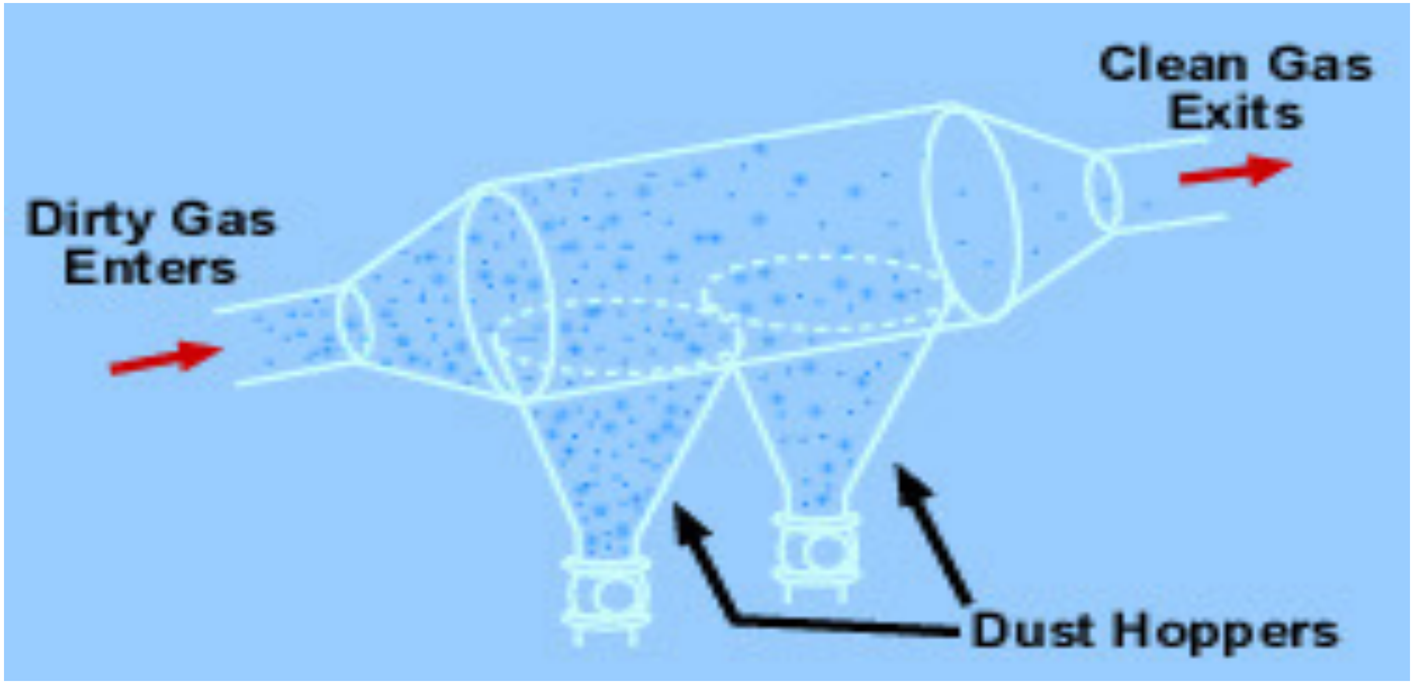
As cyclone diameter reduces, efficiency of collection increases, so does the pressure drop, which implies an increase in power requirement to move the gas thru the collector.

Therefore to increase efficiency without increasing power consumption, multiple cyclones are used in parallel - **multiclones**



Gravity Settling Chambers

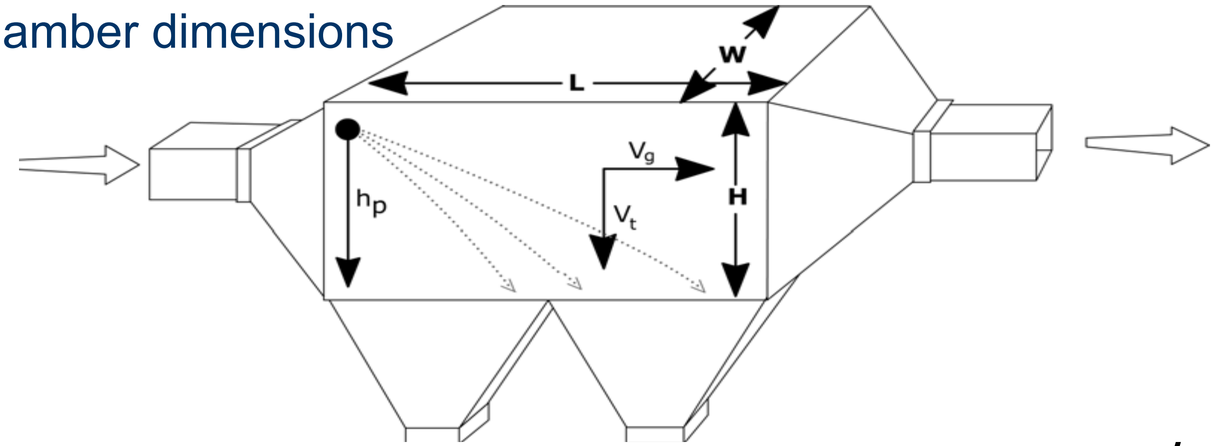
Settling chambers use gravity to separate solid particles. The gas stream enters a chamber with large cross-sectional area perpendicular to the gas flow so that the gas velocity is reduced, and large particles drop out of the gas and are collected in hoppers.



Because settling chambers are effective in removing only larger particles, they are used in conjunction with a more efficient control device.

Gravity Settling Chambers - contd

Settling chamber dimensions



The efficiency of a settling chamber can be estimated by: $\eta = \dfrac{v_t L}{H v_g}$

Where:

v_g = gas stream velocity,

v_t = terminal settling velocity,

h_p = height particle enters the chamber

The diameter of the smallest particle that can be completely removed (corresponds to η = 1) is given by:

$$D_{p,\min} = \sqrt{\frac{18 \mu v_t}{g(\rho_p - \rho)}}$$ (Stoke's law)

Gravity Settling Chambers - contd

The expression for $D_{p,\min}$ can be rewritten in so many ways:

$$D_{p,\min} = \sqrt{\frac{18\mu H v_g}{gL(\rho_p - \rho)}} \quad \text{or} \quad D_{p,\min} = \sqrt{\frac{18\mu Q}{gWL(\rho_p - \rho)}}$$

The design variables for a settling chamber include the length, width and height of the chamber. These parameters are chosen such that all particles above a specified size will be removed.

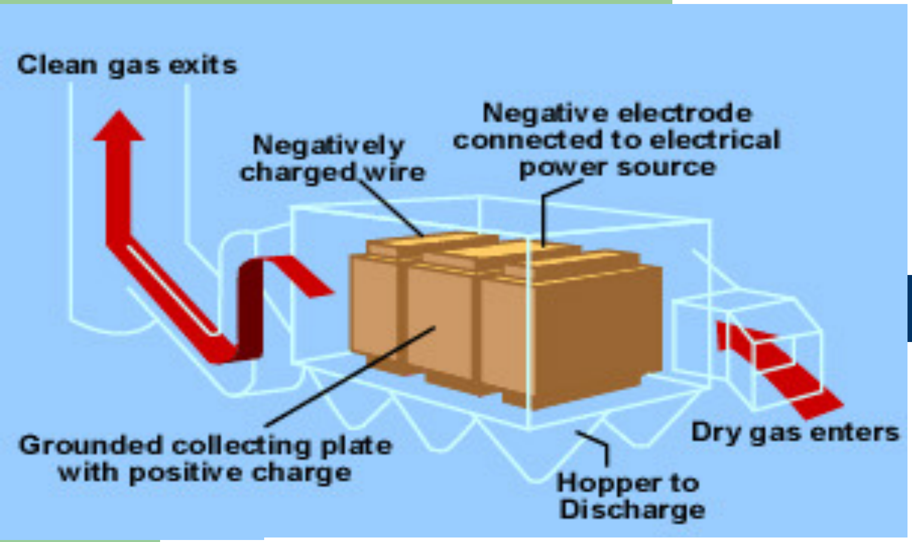
Advantages:

- Low Capital Cost
- Very Low Energy Cost
- No Moving Parts
- Few Maintenance Requirements
- Low Operating Costs
- Excellent Reliability
- Low Pressure Drop
- Device Not Subject to Abrasion
- Provides Incidental Cooling of Gas Stream
- Dry Collection and Disposal

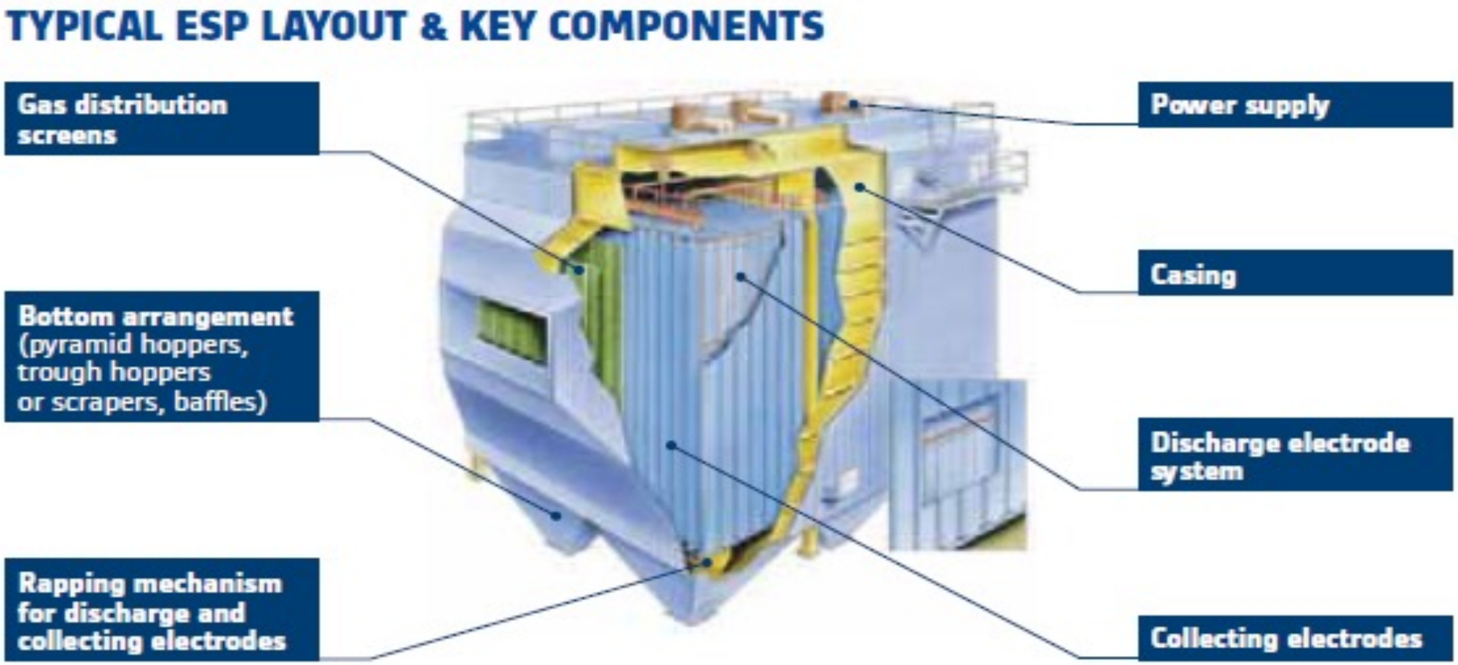
Disadvantages:

- Relatively Low Collection Efficiencies
- Large Physical Size – severe limitation

Electrostatic Precipitators (ESPs)



- Mostly used for flue gases
- Removal efficiency is about 99 percent.



Steps involved in the ESP process

- Application of high voltage to a pair of electrodes, one of which has a small radius capable of causing a high electric field – corona - in the surrounding gas, thereby negatively charging the particles.
- Particles migrate to the collector, where they get neutralized
- Collected particles are removed by rapping and drop to the hopper below

2 types of ESP

- One- stage – where ionization and collection are achieved in a single stage
  - Tubular (wire-in- tube)
  - Flat (wire-in- plate)
- two- stage – where ionization in one stage followed by particle collection in the second stage stage

Collection efficiency for ESPs is given by the Deutsch-Anderson equation:

$$\eta = 1 - e^{-Av/Q}$$

Where A = collector surface area
v = particle drift (migration) velocity (see next slide)
Q = gas volumetric flowrate

One important factor that affects actual collection efficiency is the particle resistivity – i.e. resistance to electrical conduction

Particles with low resistivity **(10^4 – 10^7 ohm.cm)** easily become charged and readily release their charge to the grounded collection plate - more likely to be re- entrained

Particles that exhibit high resistivity **(> 10^{10} ohm.cm)** are difficult to charge, but once charged, they do not readily give up their acquired charge on arrival at the collection - difficult to dislodge

Most ESPs are designed using a particle-migration velocity based on field experience rather than theory. Typical particle migration velocity rates, published by various ESP vendors is shown below:

Typical effective particle-migration velocity rates for various applications

Application	Migration velocity (ft/sec)	Migration velocity (cm/s)
Utility fly ash	0.13-0.67	4.0-20.4
Pulverized coal fly ash	0.33-0.44	10.1-13.4
Pulp and paper mills	0.21-0.31	6.4-9.5
Sulfuric acid mist	0.19-0.25	5.8-7.62
Cement (wet process)	0.33-0.37	10.1-11.3
Cement (dry process)	0.19-0.23	6.4-7.0
Gypsum	0.52-0.64	15.8-19.5
Smelter	0.06	1.8
Open-hearth furnace	0.16-0.19	4.9-5.8
Blast furnace	0.20-0.46	6.1-14.0
Hot phosphorous	0.09	2.7
Flash roaster	0.25	7.6
Multiple-hearth roaster	0.26	7.9
Catalyst dust	0.25	7.6
Cupola	0.10-0.12	3.0-3.7

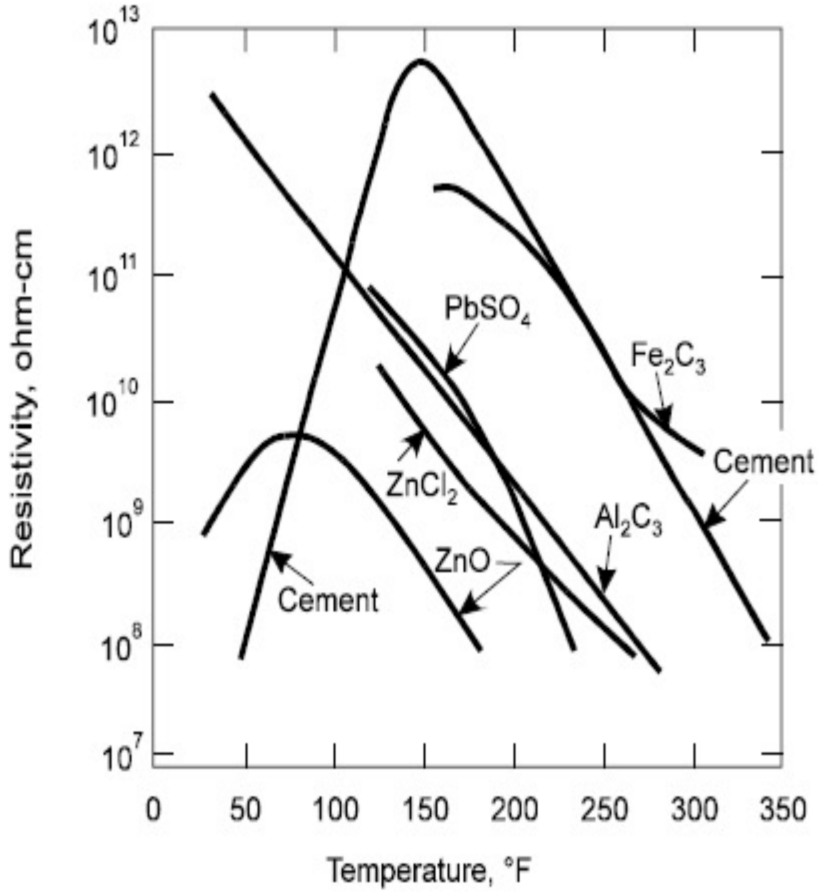


Fig 1

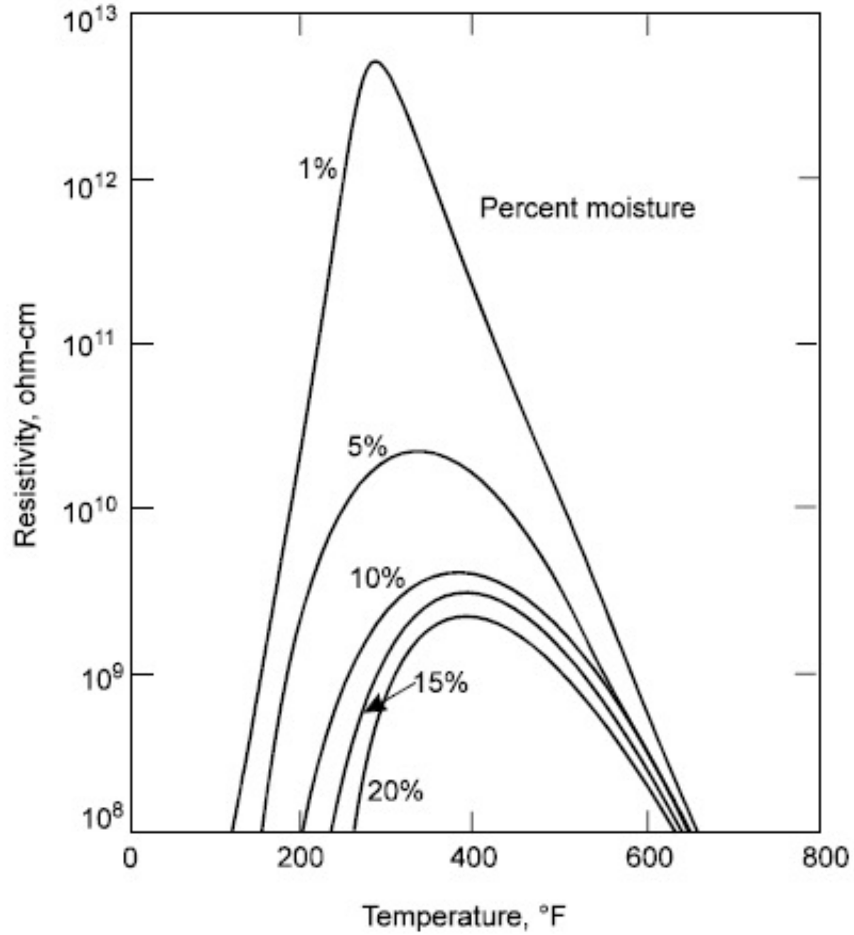


Fig 2

Figure 1 shows the variation in resistivity with changing gas temperature for six different industrial dusts while figure 2 shows the effect of temperature and moisture on the resistivity of cement dust

Advantages and Disadvantages of an electrostatic precipitator

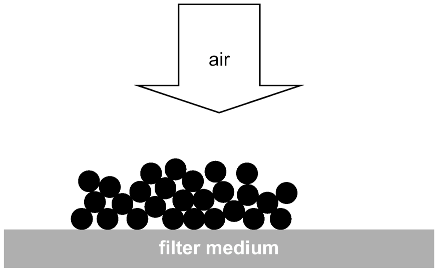
Advantages

- Electrostatic precipitators are capable very high efficiency, generally of the order of 99.5-99.9%.
- Since the electrostatic precipitators act on the particles and not on the air, they can handle higher loads with lower pressure drops.
- They can operate at higher temperatures.
- The operating costs are generally low.

Disadvantages

- Initial cost is highest for any particulate collection system
- Although they can be designed for a variety of operating conditions, they are not very flexible to changes in the operating conditions, once installed.
- Large space required for installation
- Dependent upon resistivity of PM, cannot be used around explosive gases

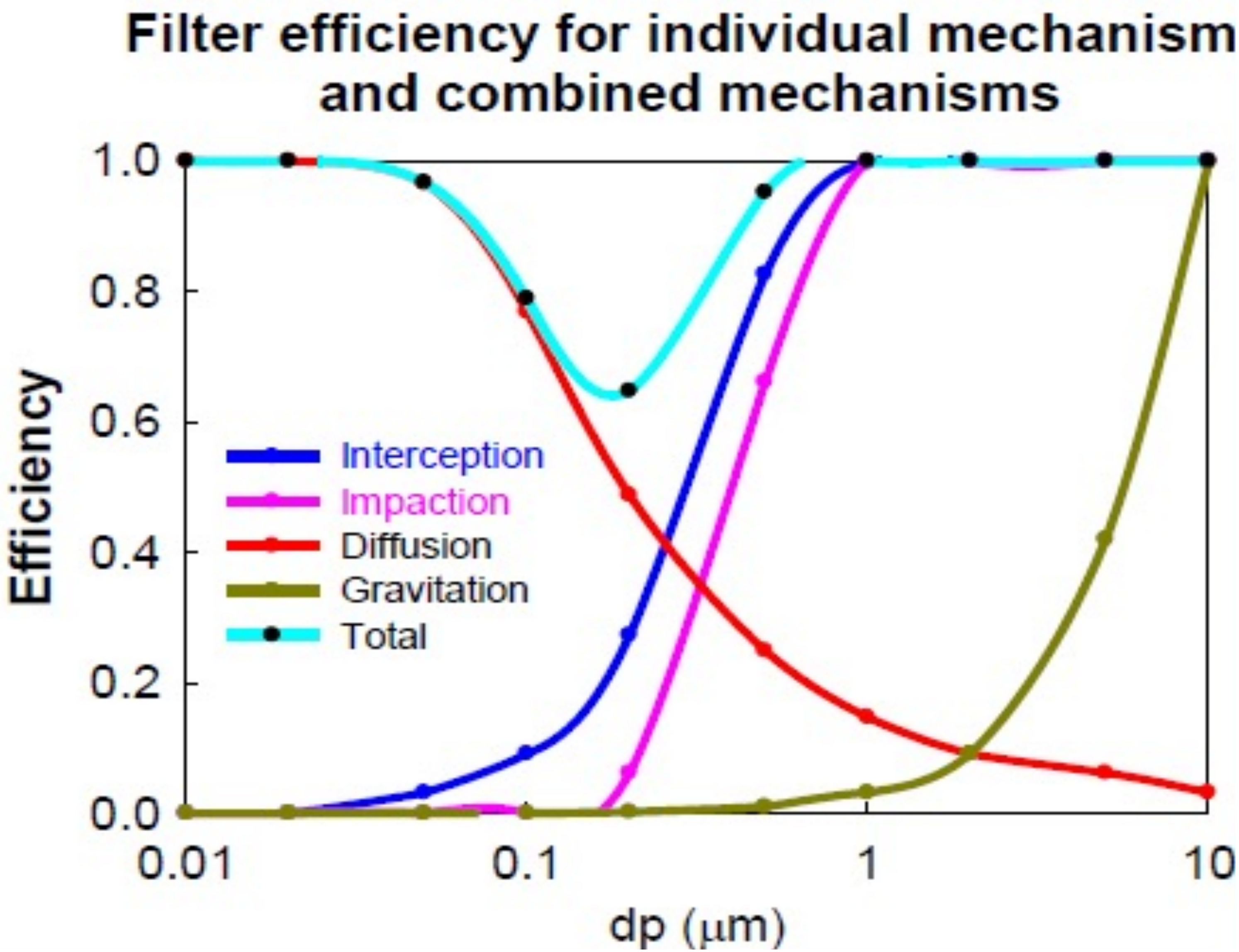
Fabric Filters (or baghouses)



Collect particulate matter on the surfaces of filter bags by

- inertial impaction,
- interception,
- Brownian diffusion, and
- sieving on already collected particles that have formed a dust layer on the bags.
- The fabric material can also capture particles that have penetrated through the dust layers.
- Electrostatic attraction may also contribute to particle capture in the dust layer and in the fabric itself.

Due to the multiple mechanisms of particle capture possible, fabric filters can be highly efficient for the entire particle size range of interest in air pollution control.


Filter efficiency for individual mechanism and combined mechanisms
1.0
0.8
0.6
0.4
0.2
0.0
Efficiency
Interception
Impaction
Diffusion
Gravitation
Total
0.01
0.1
1
10
dp (μm)

Fabric Filters- Contd

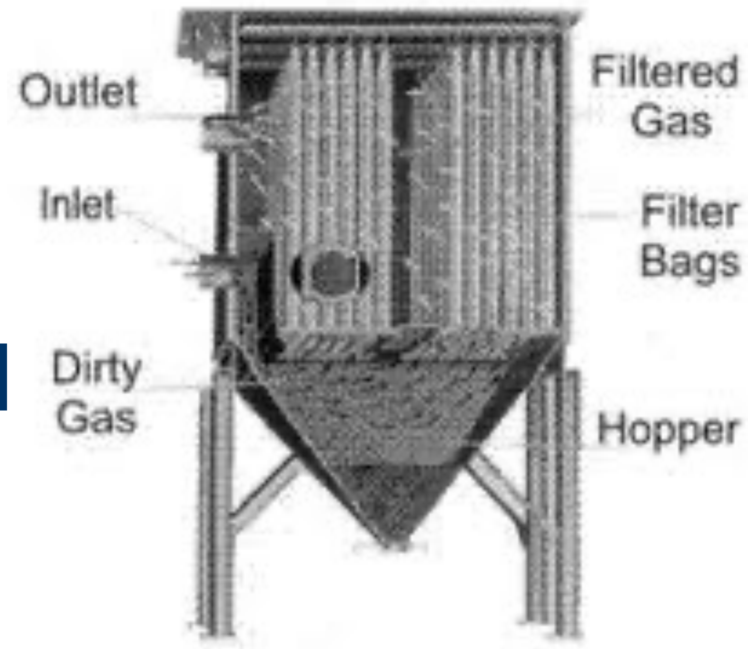
Filter Medium

The filter medium can be cloth, metal, glass, plastic etc. The most common being cloth, which can be of any material and weave selected to fit the specific application, considering

- composition of the gas,
- operating temperature,
- dust loading, and
- the physical and chemical characteristics of the particulate.

Filter bags are usually tubular, 0.1 – 0.35m in diameter and 2 – 10m in length. They are usually arranged in groups in separate compartments

Types of fabric filter (Baghouses)



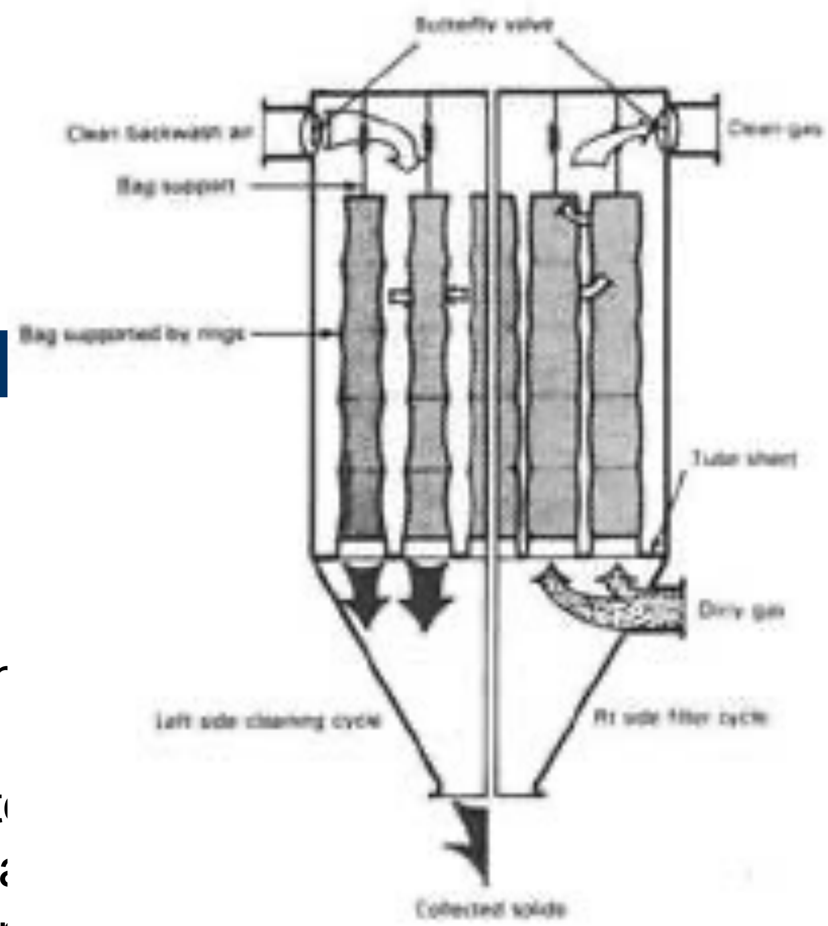
Dust enters the baghouse compartment through inlet on the hoppers. Larger particles drop out while smaller dust particles collect on filter bags. When the dust layer thickness reaches a level where flow through the system is sufficiently restricted (determined by the *pressure drop*), bag cleaning is initiated. Cleaning can be done while the baghouse is still online (filtering) or in isolation (offline). Once cleaned, the compartment is placed back in service and the filtering process starts over.

The three common types of baghouses based on their cleaning methods are:

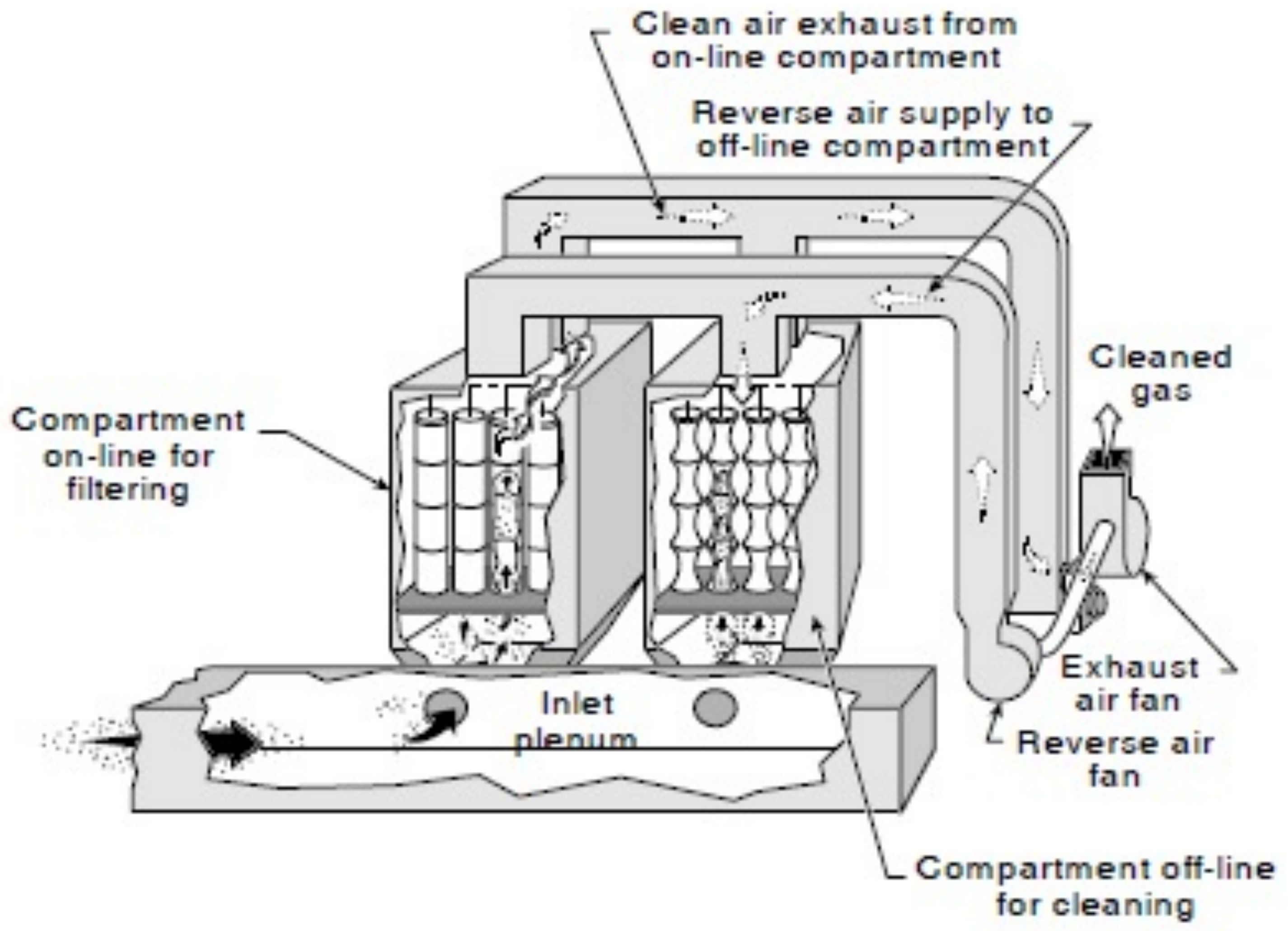
- Reverse-air
- Shaker
- Pulse-jet

Reverse air baghouse

Gas flow to the bags is stopped in th
compartment being cleaned and
reverse (outside-in) air flow is direct
through the bags. This reversal of ga
flow gently collapses the bags towar
their centerlines, which causes the
cake to detach from the fabric surface.



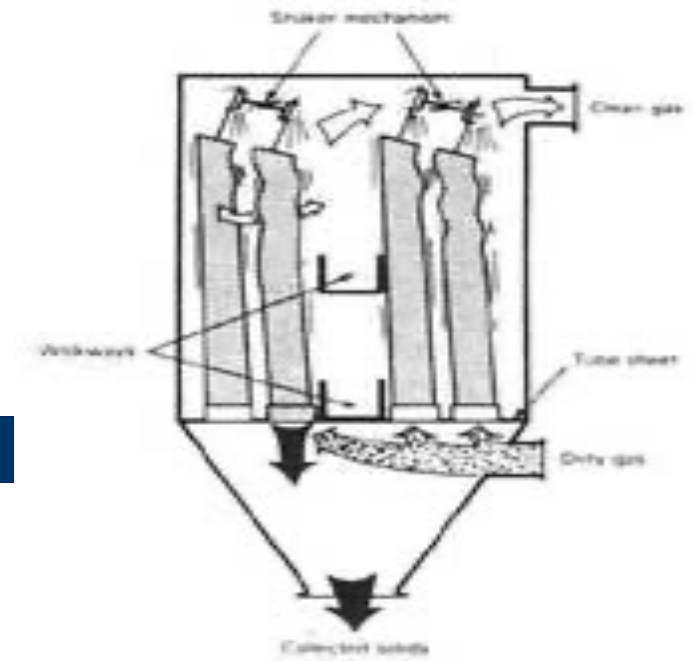
Reverse-Air Baghouse



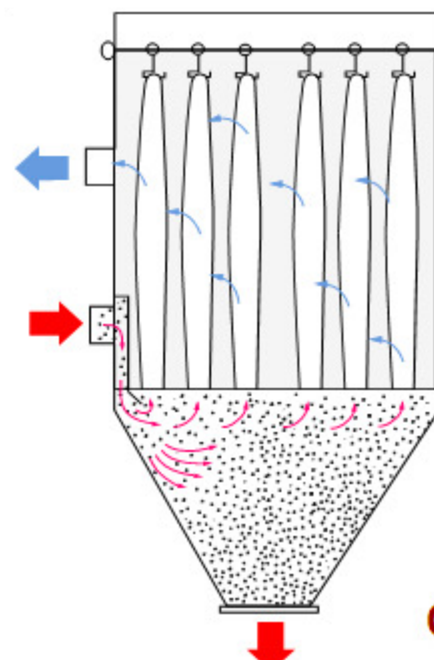
Typical Reverse-Air Baghouse

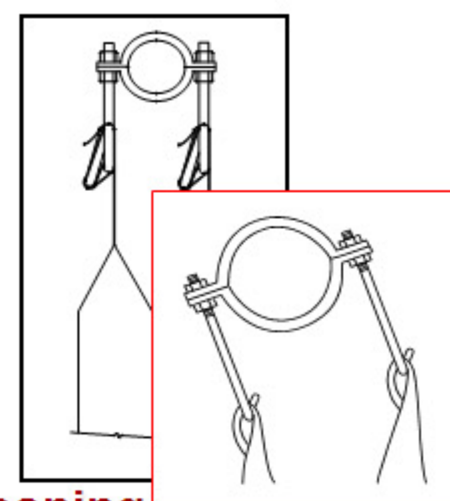
Shaker baghouse

This cleaning method physically shakes the bags in order to mechanically release the dust cake. The tubular filter bags are fastened onto a cell plate at the bottom of the baghouse and suspended from horizontal beams at the top.



Mechanical-Shaker Baghouse

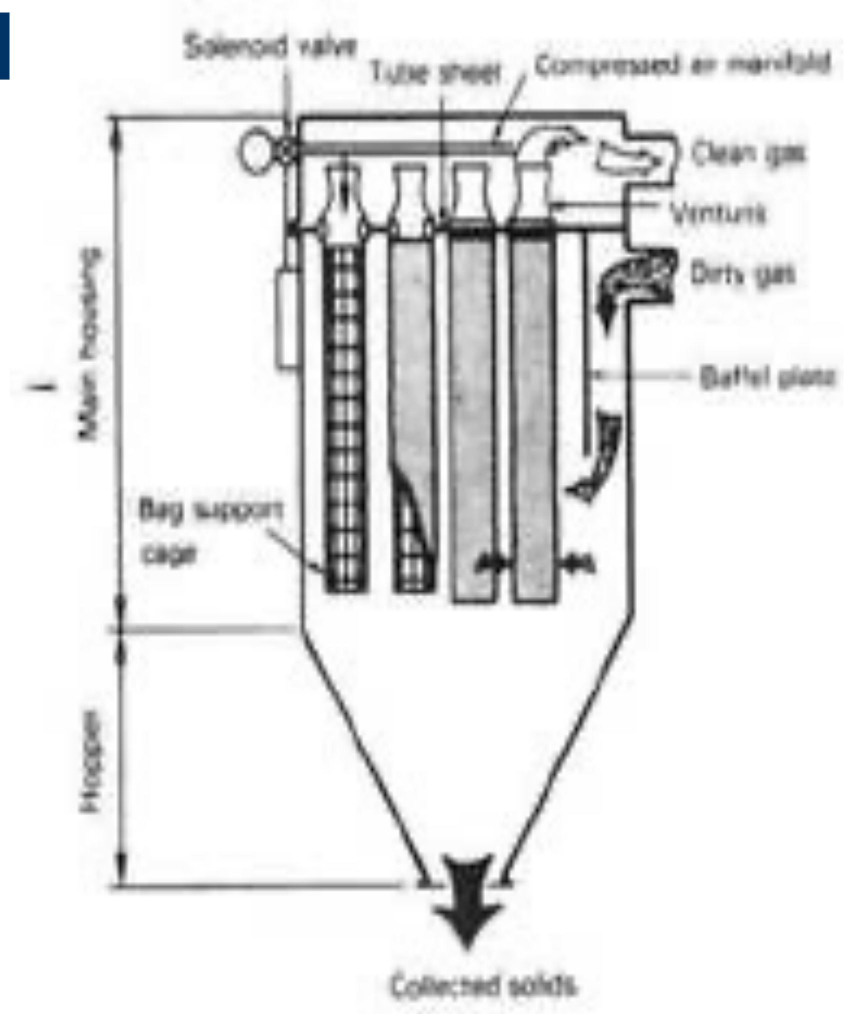




Cleaning

Pulse jet baghouse

Here the bags are supported on metal wire cages that are suspended from the top of the unit. Particulate-laden gas flows around the *outside* of the bags, and a dust cake accumulates on the exterior surfaces. When cleaning is needed, a very-short-duration pulse of compressed air is injected at the top inside part of each bag in the row of bags being cleaned. The compressed air pulse generates a pressure wave that moves down each bag and, in the process, dislodges some of the dust cake from the bag. The metal cage prevents collapse of the bag



Advantages and Disadvantages of the Fabric filter system

Advantages of Fabric Filters

- Very high collection efficiency
- They can operate over a wide range of volumetric flow rates
- The pressure drops are reasonably low.
- Filtered air can be recirculated
- Fabric Filter houses are modular in design, and can be pre-assembled at the factory
- Fabric filters are available in a large number of configurations

Disadvantages of Fabric Filters

- Fabric Filters require a large floor area
- The fabric is damaged at high temperature.
- Ordinary fabrics cannot handle corrosive gases.
- Fabric Filters cannot handle moist gas streams
- A fabric filtration unit is a potential fire hazard
- Replacement of fabric is a respiratory hazard – respiratory protection required

Successful design of a fabric filter depends on key design variables

- Filter bag material
- Fabric cleaning method
- Air-to-cloth ratio
- Baghouse configuration (i.e., forced or induced draft)
- Materials of construction

Design considerations

- **Selection of fabric** – this is determined on the basis of operating temperature, characteristics of the gas stream etc. (see below)

Characteristics of Several Fibers Used in Fabric Filtration

Fiber type	Max operating temp. (°F)	Resistance[b] Abrasion	Mineral acids	Organic acids	Alkalis	Solvent
Cotton[c]	180	VG	P	G	P	E
Wool[d]	200	F/G	VG	VG	P/F	G
Modacrylic[d] (Dynel™)	160	F/G	E	E	E	E
Polypropylene[d]	200	E	E	E	E	G
Nylon polyamide[d] (Nylon 6 and 66)	200	E[e]	F	F	E	E
Acrylic[d] (Orlon™)	260	G	VG	G	F/G	E
Polyester[d] (Dacron[f])	275	VG	G	G	G	E
(Creslan™)	250	VG	G	G	G	E
Nylon aromatic[d] (Nomex™)	375	E	F	G	E	E
Fluorocarbon[d] (Teflon™, TFE)	450	F/G	E[g]	E[g]	E[g]	E[g]
Fiberglass[c]	500	F/G[h]	G	G	G	E
Ceramics[i] (Nextel 312™)	900+	—	—	—	—	—

- The most important design parameter is the **air –to-cloth ratio** aka filtering velocity – volumetric flow rate / surface area of fabric

After a fabric has been selected, an initial gas-to-cloth ratio can be determined using the manufacturer's guideline

Air-to-Cloth Ratios

Dust	Shaker/woven Reverse-air/woven	Pulse jet/felt
Alumina	2.5	8
Asbestos	3.0	10
Bauxite	2.5	8
Carbon black	1.5	5
Coal	2.5	8
Cocoa, chocolate	2.5	12
Clay	2.5	9
Cement	2.0	8
Cosmetics	1.5	10
Enamel frit	2.5	9
Feeds, grain	3.5	14
Feldspar	2.2	9
Fertilizer	3.0	8
Flour	3.0	12

Bag Prices

Type of cleaning	Bag diameter (in.)	Type of material[a]						
		PE	PP	NO	HA	FG	CO	TF
Pulse jet, TR[b]	4½ –5⅛	0.59	0.61	1.88	0.92	1.29	NA[c]	9.05
	6–8	0.43	0.44	1.56	0.71	1.08	NA	6.80
Pulse jet, BBR	4½–5⅛	0.37	0.40	1.37	0.66	1.24	NA	8.78
	6–8	0.32	0.33	1.18	0.58	0.95	NA	6.71
Shaker								
Strap top	5	0.45	0.48	1.28	0.75	NA	0.44	NA
Loop top	5	0.43	0.45	1.17	0.66	NA	0.39	NA
Reverse air with rings	8	0.46	NA	1.72	NA	0.99	NA	NA
Reverse air	8	0.32	NA	1.20	NA	0.69	NA	NA
w/o rings[d]	11½	0.32	NA	1.16	NA	0.53	NA	NA

Note: For pulse-jet baghouses, all bags are felts except for the fiberglass, which is woven. For bottom access pulse jets, the cage price for one cage can be calculated from the single-bag fabric area using the following:

In 50 cage lots:	$\$ = 4.941 + 0.163\ ft^2$	$\$ = 23.335 + 0.280\ ft^2$
In 100 cage lots:	$\$ = 4.441 + 0.163\ ft^2$	$\$ = 21.791 + 0.263\ ft^2$
In 500 cage lots:	$\$ = 3.941 + 0.163\ ft^2$	$\$ = 20.564 + 0.248\ ft^2$

*a*PE = 16-oz polyester; PP = 16-oz polypropylene; NO = 14-oz nomex; HA = 16-oz homopolymer acrylic; FG = 16-oz fiberglass with 10% TeflonTM; CO = 9-oz cotton; TF = 22-oz TeflonTM felt.
*b*Bag removal methods: TR = top bag removal (snap in); BBR = bottom bag removal
*c*NA = Not applicable
*d*Identified as reverse-air bags, but used in low-pressure pulse applications.

Precooling of gases

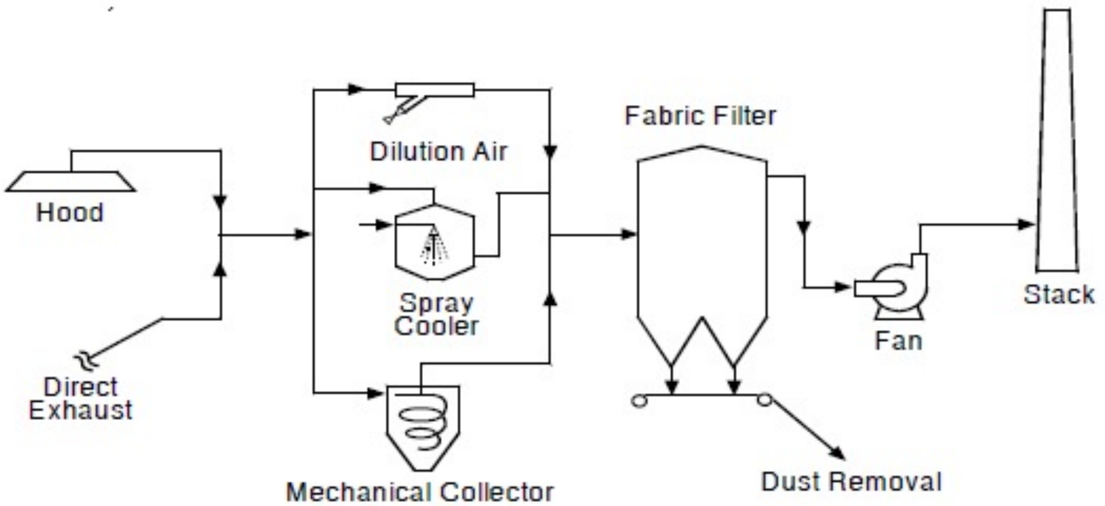
three precooling methods are:

- Radiation and convection – use of heat exchangers
- Spray cooling with water – cooled gas stream must be well above the dew point to prevent condensation and plugging of the bags. Quantity of water vapourised must be used to determine total gas volume to be filtered
- Addition of outside air – increases baghouse size and fan load

Baghouse configuration

Baghouses have two basic configurations,

- gases either pushed through the system by a fan located on the upstream side (forced draft fan) or
- pulled through by a fan on the downstream side (induced draft fan)



- ## Construction Materials

The most common material used in fabric-filter construction is carbon steel.

In special cases (eg acidic gases) , stainless steel may be required. Stainless steel will increase the cost of the fabric filter significantly when compared to carbon steel.

SELECTING A FILTER BAG SYSTEM

It is proposed to install a pulse-jet fabric-filter system to remove particulates from an air stream. Select the most appropriate bag from the four proposed below. The volumetric flow rate of the air stream is 10,000 std ft^3/min ($4.72\ m^3/s$) (standard conditions being 60°F and 1 atm), the operating temperature is 250°F (394 K), the concentration of pollutants is 4 $grains/ft^3$ (141 $grains/m^3$), the average air-to-cloth ratio is $(2.5\ ft^3/min)/ft^2$, and the required collection efficiency is 99 percent.

Information on the four proposed bags is as follows:

Bag designation	A	B	C	D
Tensile strength	Excellent	Above average	Fair	Excellent
Recommended maximum temperature, °F	260	275	260	220
Cost per bag	$26.00	$38.00	$10.00	$20.00
Standard size	8 in by 16 ft	10 in by 16 ft	1 ft by 16 ft	1 ft by 20 ft

Note: No bag has an advantage from the standpoint of durability.

Procedure

1. ***Eliminate from consideration any bags that are patently unsatisfactory.*** Bag D is eliminated because its recommended maximum temperature is below the operating temperature for this application. Bag C is also eliminated, because a pulse-jet fabric-filter system requires that the tensile strength of the bag be at least above average.

2. ***Convert the given flow rate to actual cubic feet per minute.*** The flow rate as stated corresponds to flow at 60°F, whereas the actual flow q will be at 250°F. Accordingly,

$$q = (10{,}000)(250 + 460)/(60 + 460) = 13{,}654 \text{ actual ft}^3\text{/min } (6.44 \text{ actual m}^3\text{/s})$$

3. ***Establish the filtering velocity v_f.*** The air-to-cloth ratio is (2.5 ft³/min)/ft². Dimensional simplification converts this to 2.5 ft/min (0.0127 m/s).

4. ***Calculate the total filtering area required.*** This equals the actual volumetric flow rate (from step 2) divided by the filtering velocity (from step 3). Thus,

$$\text{Total filtering area} = 13{,}654/2.5 = 5461.6 \text{ ft}^2 (507.9 \text{ m}^2)$$

5. ***Calculate the filtering area available per bag.*** Consider the operating bag to be in the form of a cylinder, whose wall constitutes the filtering area. This is accordingly calculated from the formula $A = \pi Dh$, where A is area, D is bag diameter, and h is bag height. The two bags still under consideration are Bag A and Bag B. The area of each is as follows:

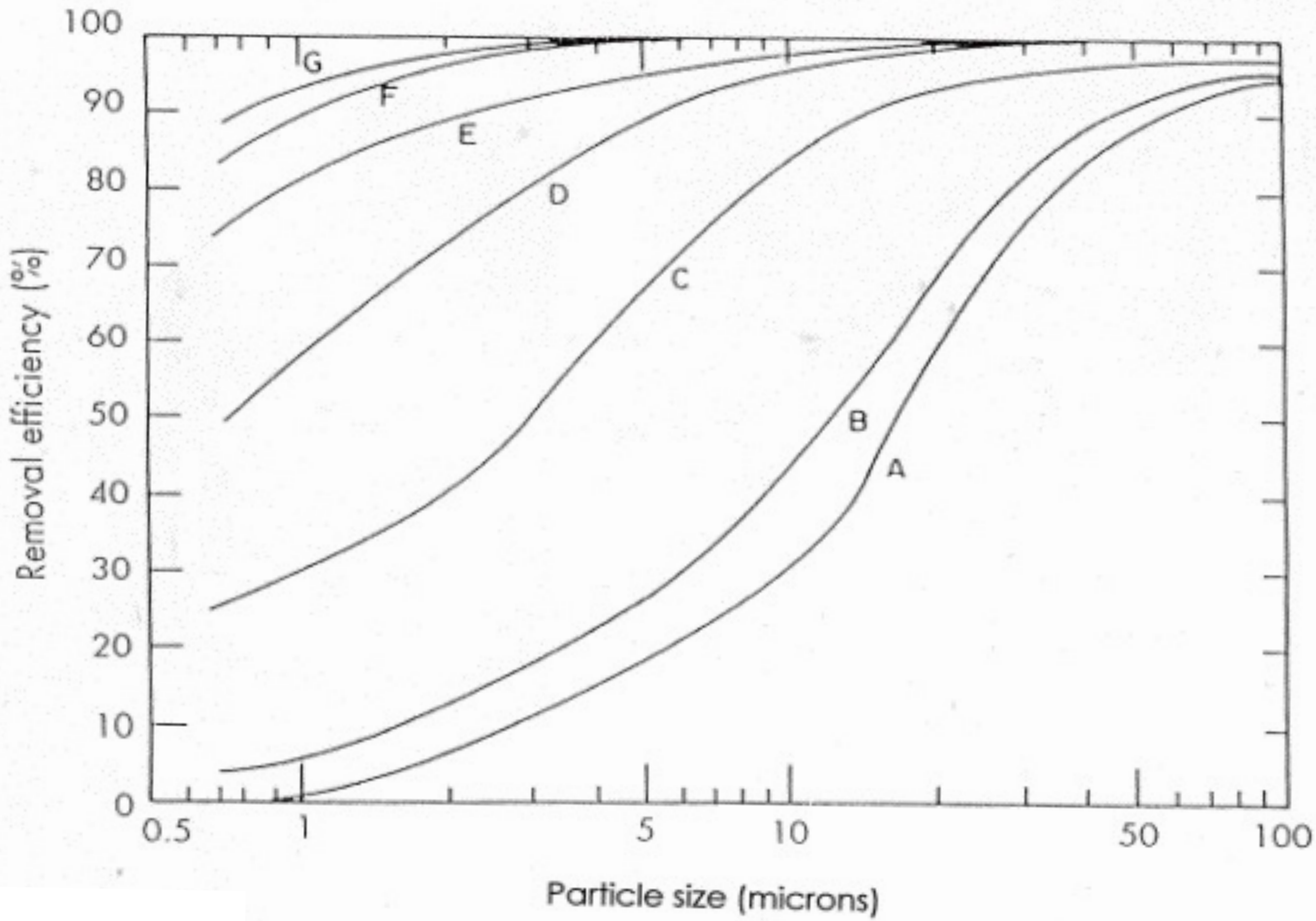
For Bag A, $A = \pi(8/12)(16) = 33.5 \text{ ft}^2 \ (3.12 \text{ m}^2)$

For Bag B, $A = \pi(10/12)(16) = 41.9 \text{ ft}^2 \ (3.90 \text{ m}^2)$

6. ***Determine the number of bags required.*** The total filtering area required is 5461.6 ft². Accordingly, if Bag A is selected, the number of bags needed is 5461.6/33.5, i.e., 163 bags. If instead Bag B is selected, the number needed is 5461.6/41.9, i.e., 130 bags.

7. ***Determine the total bag cost.*** If Bag A is used, the total cost is (163)($26.00), i.e., $4238. If instead Bag B is used, the total cost is (130)($38.00), i.e., $4940.

8. ***Select the most appropriate bag.*** Since the total cost for Bag A is less than that for Bag B, select Bag A.



Comparison of removal efficiencies: (A) baffled settling chamber, (B) cyclone "off the shelf," (C) carefully designed cyclone, (D) electrostatic precipitator, (E) spray tower, (F) venturi scrubber, (G) bag filter

General guideline to choosing a particulate control system